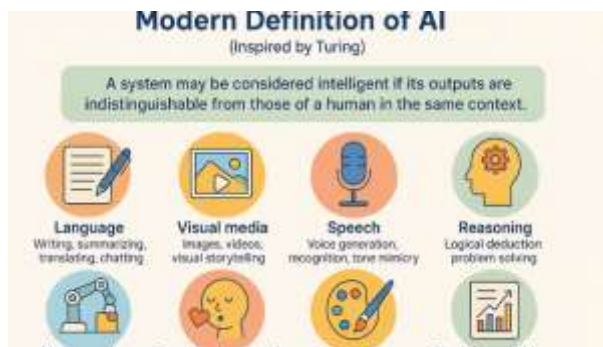


AI CONCEPTS

Q1. What is Artificial Intelligence?

Artificial Intelligence is a technology that builds systems capable of performing tasks that which human can perform, such as Learning, Reasoning, Problem-solving and Decision-making.

AI can work faster than the humans. It enables to process the large datasets and recognize patterns and operate it.



Core Concepts:

Machine Learning - It is branch of AI where computers learn from data and past experiences with step by step instructions.

Generative AI - It is a type of AI that can create brand new content like text, images, music or code rather than just analysing existing data.

Natural Language Processing - It is a branch of AI that gives computers the ability to read, understand, and generate human language.

Common Applications:

Healthcare

Finance

Customer Service

Transportation

Q2. Difference between AI, Machine learning, Generative AI, and Agentic AI?

Artificial Intelligence (AI)

AI is the broadest concept. It refers to machines or software that perform tasks requiring human-like intelligence, such as reasoning, decision-making, perception, language understanding, or problem-solving.

Examples:

- Spam email filters
- Chess-playing programs
- Voice assistants
- Self-driving car systems



Machine Learning (ML)

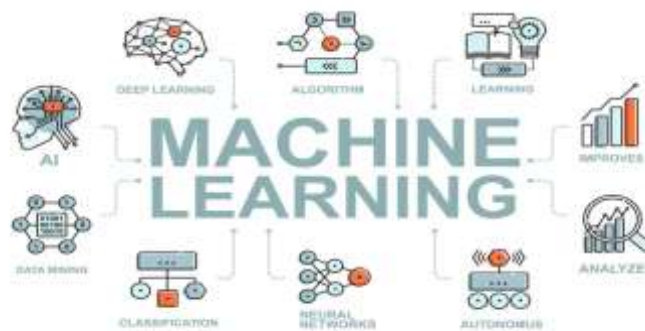
ML is subset of AI.

Instead of being explicitly programmed with every rule, ML systems learn patterns from data and use those patterns to make predictions or decisions.

Examples:

- Predicting house prices from historical data
- Recommending movies on streaming platforms

- Fraud detection in banking
- Image recognition



Generative AI (GenAI)

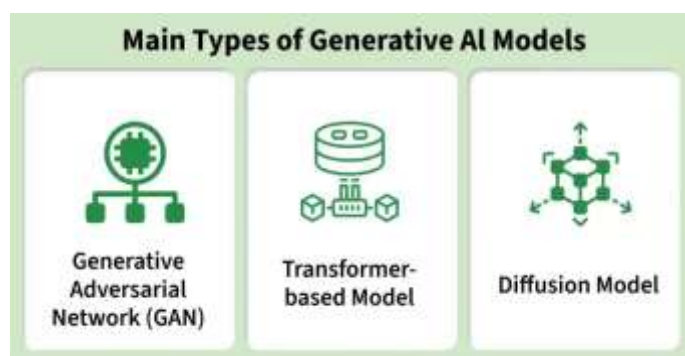
Generative AI is a subset of Machine Learning focused on creating new content.

Rather than only classifying or predicting, it generates:

- Text
- Images
- Music

Examples:

- [ChatGPT](#) generating text
- [Claude](#) answering questions
- [Midjourney](#) creating images
- [GitHub Copilot](#) generating code



Agentic AI

Agentic AI goes beyond generating content. It can pursue goals, make decisions, use tools, and take multiple actions autonomously.

An AI agent typically:

1. Understands a goal.
2. Creates a plan.
3. Uses tools (search, databases, APIs, software).
4. Evaluates results.
5. Adjusts its approach.
6. Continues until the goal is achieved.

Example: Goal: "Book me the cheapest flight to Mumbai next weekend and send me the options."

An Agentic AI system might:

- Search flight websites.
- Compare price
- Filter by your preferences
- Prepare options
- Potentially complete the booking process



Q3. Explain any three open-source AI models available on hugging face?

Three widely used open-source AI models available on Hugging Face

1. Llama 3

Llama 3 is a family of large language models designed for tasks such as text generation, summarization, question answering, coding assistance, and reasoning.

Key Features

- Available in multiple parameter sizes (e.g., 8B and 70B versions).
- Strong performance on language understanding and generation tasks.
- Open-weight model that researchers and developers can deploy locally.

Applications

- Chatbots and virtual assistants
- Content generation
- Document summarization

Advantages

- High-quality text generation
- Flexible deployment options

2. Mistral 7B

Mistral 7B is a compact yet powerful language model known for delivering strong performance relative to its size.

Key Features

- Approximately 7 billion parameters.
- Efficient architecture that requires fewer computational resources.
- Suitable for running on consumer-grade hardware.

Applications

- Customer support chatbots
- Educational assistants
- Local AI deployments

Advantages

- Lower hardware requirements
- Fast inference speed

3. Stable Diffusion

Stable Diffusion is an image-generation model that creates images from text descriptions.

Key Features

- Converts natural language prompts into images.
- Supports image editing and image-to-image generation
- Runs locally on many modern GPUs.

Applications

- Digital art creations
- Graphic design
- Marketing content generation

Advantages

- High-quality image generation
- Extensive customization options



Q4. What is the difference between traditional programming and AI- assisted development?

Traditional programming and AI-assisted development differ mainly in how software is created and how solutions are produced.

Aspect	Traditional Programming	AI-Assisted Development
Code creation	Written manually	AI suggests or generates code
Development speed	Slower	Faster
Productivity	Depends entirely on developer	Enhanced by AI assistance
Learning curve	Requires detailed knowledge	AI can help explain concepts
Debugging	Manual	AI can suggest fixes
Documentation	Written manually	AI can generate automatically
Human involvement	100% coding effort	Human supervises and validates

Q5. Explain AI Hallucinations with one example?

AI Hallucinations

An **AI hallucination** happens when an AI system gives information that sounds correct and confident but is actually wrong, made up, or unsupported by facts.

Simple Example

If someone asks : Who invented the smartphone in 2015?

Then AI replies : Raju invented the smartphone in 2015.

This answer is a hallucination because:

- There is no well-known inventor named "Raju" who invented the smartphone in 2015.
- Smartphones existed before 2015.
- The AI generated a plausible-sounding but incorrect answer.

So,

AI Hallucination- When AI makes up information and presents it as if it were true.
